

CS & IT ENGINEERING



Operating System

Memory Management

Lecture -6

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Recap of Previous Lecture



doubt ?

Topic

Page Replacement Policy

Topic

Optimal, LRU, FIFO Policy



Topics to be Covered



Topic

Multilevel Paging





Topic : Question

[GATE-2016]

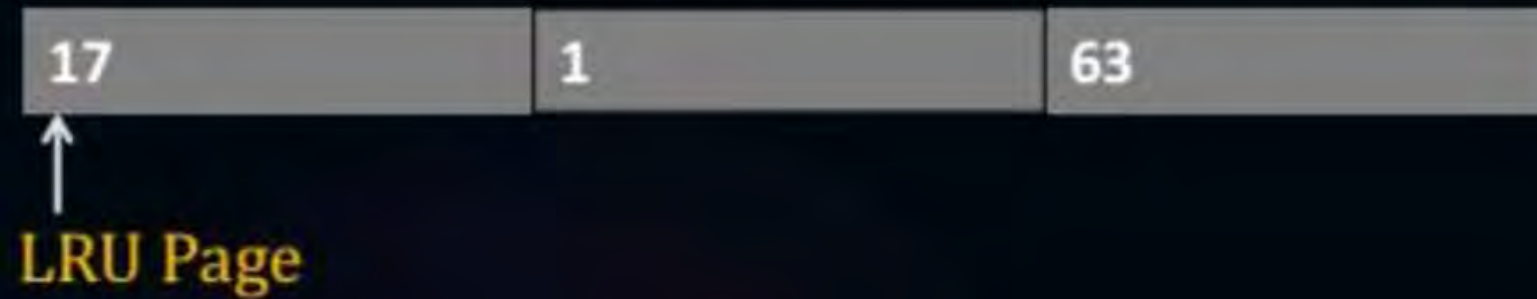


#Q. Consider a computer system with ten physical page frames. The system is provided with an access sequence $a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20}$ where each a_i is a page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is 1

$$\begin{array}{r} \text{LIFO} = 31 \\ \text{optimal} = 30 \\ \hline 1 \end{array}$$

$\left. \begin{array}{c} a_1 \\ \vdots \\ a_i \end{array} \right\}$
 ~~$a_1 a_2 a_3 \dots a_{20}$~~ $\dots$ (a_{20})

#Q. A demand paged virtual memory system uses 16 bit virtual address, page size of 256 bytes, and has 1Kbyte of main memory. LRU page replacement is implemented using the list, whose current status (page number is decimal) is



For each hexadecimal address in the address sequence given below,

00FF, 010D, 10FF, 11B0 0, 1, 16, 17

indicate

1. the new status of the list
2. page faults, if any, and $\Rightarrow 3$
3. page replacements, if any.

$$\text{no. of frames} = \frac{1KB}{256} = 4$$

17
1
63

✓
0

17

1

63

0

✗
1

✓
16

16

1

63

0

✓
17

16
1
17
0

LRU list

0	1	16	17
---	---	----	----

↑

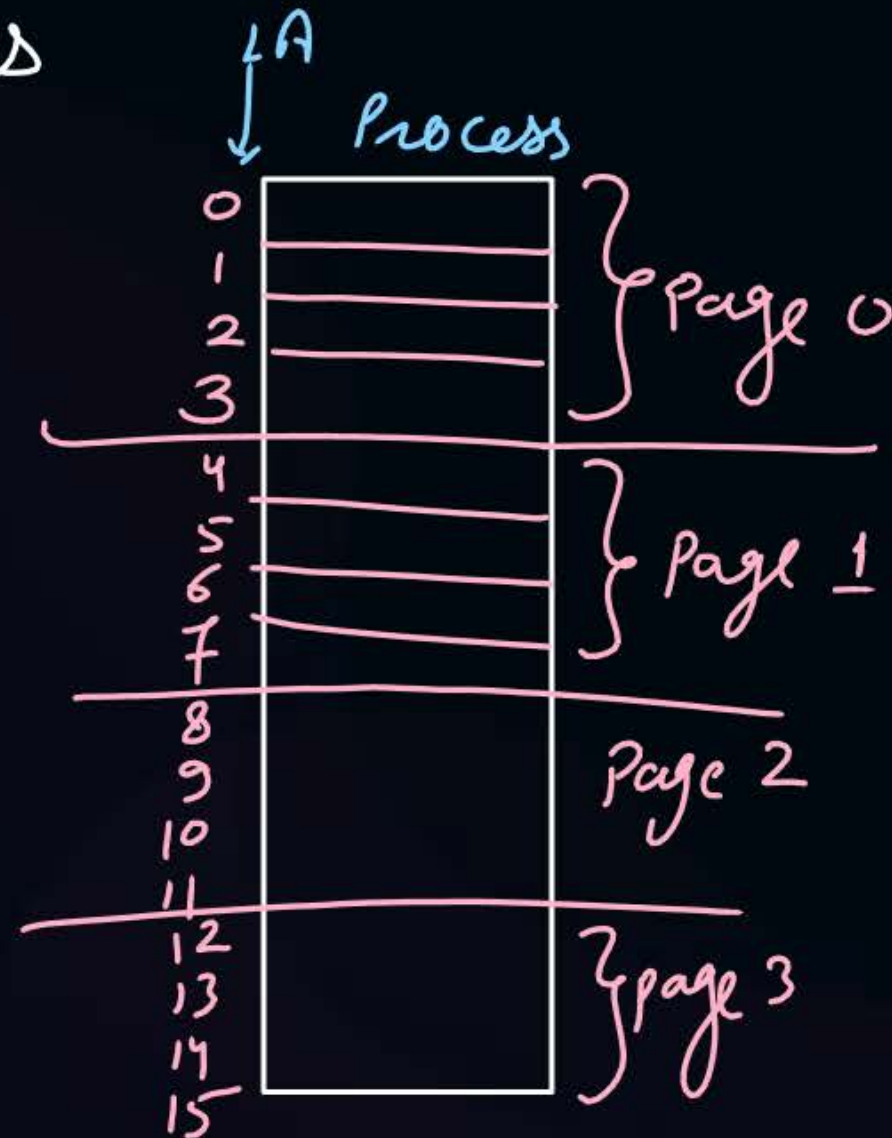
LRU



Topic : Making Page Reference Sequence

If L.A. is given in decimal

ex:- Page size = 4 bytes



$$P = \left\lfloor \frac{L.A.}{\text{Page Size}} \right\rfloor$$

$$d = L.A. \% \text{ Page Size}$$

$$14 \% 4 = 2$$

P.A. in decimal

$$f = \left\lfloor \frac{P.A.}{\text{Page size}} \right\rfloor$$

$$d = P.A. \% \text{ Page size}$$

$$L.A. = (p * \text{page size}) + d$$

$$P.A. = (f * \text{page size}) + d$$

#Q. Assume that a main memory with only 4 pages, each of 16 bytes, is initially empty. The CPU generates the following sequence of virtual addresses and uses the Least Recently Used (LRU) page replacement policy.

0, 4, 8, 20, 24, 36, 44, 12, 68, 72, 80, 84, 28, 32, 88, 92

How many page faults does this sequence cause? What are the page numbers of the pages present in the main memory at the end of the sequence?

A 6 and 1,2,3,4

B 7 and 1,2,4,5

C 8 and 1,2,4,5

D 9 and 1,2,3,5

$$P = \left\lfloor \frac{LA}{16} \right\rfloor$$

Page Ref. Seq :- 0, 1, 2, 0, 4, 5, 1, 2, 5

0	0	0	0	0	0	2
	1	1	1	5	5	5
		2	2	2	1	1
			4	4	4	4

LA to PA translation in decimal:-

$$p = \left\lfloor \frac{LA}{\text{page size}} \right\rfloor$$

$$d = LA \% \text{page size}$$

→ search
in PT

P	f

$$PA = (f * \text{page size}) + d$$

Ques) mem size = 256KB
process size = 512B
page size = 128B

$$L.A. = 493$$

$$P.A. = \underline{3181} ?$$

	P.T.
0	10
1	19
2	6
3	24

$$P = \left\lfloor \frac{493}{128} \right\rfloor = 3$$

$$d = 493 \% 128 = 109$$

$$P.T \Rightarrow \text{for } P=3 \Rightarrow f=24$$

$$\begin{aligned} P.A. &= (24 * 128) + 109 \\ &= \underline{\underline{3181}} \end{aligned}$$

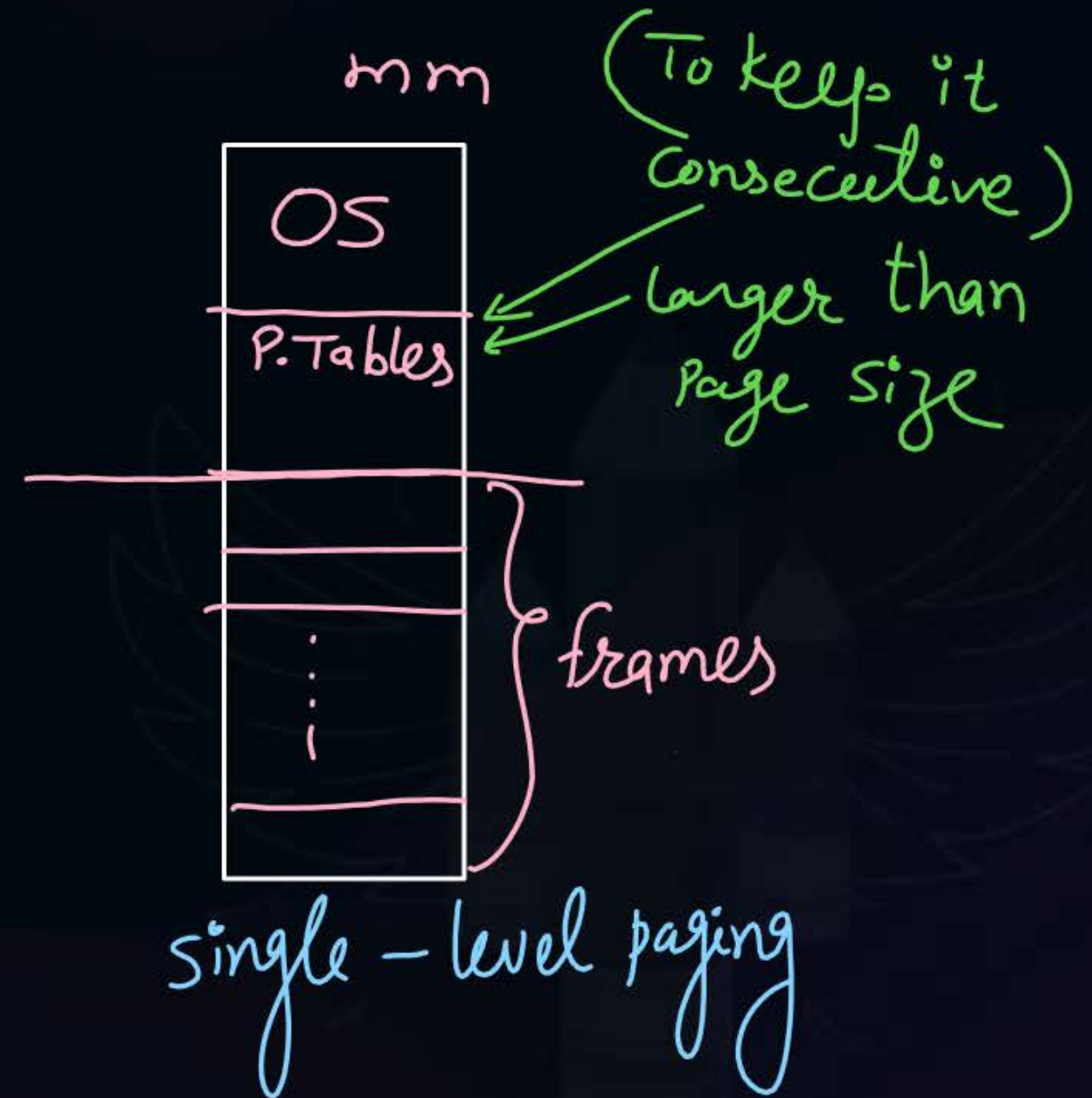


Topic : Let's Take a Simple Example

$$\left. \begin{array}{l} \text{Process Size} = 32\text{B} \\ \text{Page size} = 4\text{B} \end{array} \right\} \text{no. of pages} = \frac{32\text{B}}{4\text{B}} = 8$$

Page table entry size = 1B

$$\begin{aligned} \text{Page Table Size} &= ? = 8 * 1\text{B} \\ &= 8\text{B} \end{aligned}$$





Topic : Now What?



Can you keep entire page table in single page? $\Rightarrow$ ^{or} ~~no~~
 frame

If not then how to access the specific entry?

$$\text{no. of pages in PT} = \frac{8B}{4B} = 2$$

$$\begin{aligned} \text{no. of P.T. entries per page} &= \frac{4B}{1B} \\ &= 4 \end{aligned}$$

Then store P.T. pages also on different frames and maintain a Page table for pages of Page table.



Topic : Page Table in Memory

Process
Page 0
Page 1
Page 2
Page 3
Page 4
Page 5
Page 6
Page 7

Page no.	Page Table
000	0100
001	0111
010	1101
011	1011
100	1001
101	0010
110	1110
111	0110

Handwritten notes:
- Page 0 of P.T. (points to rows 000-011)
- Page 1 of P.T. (points to rows 100-111)
- PT of PT (points to the entire table)
- (outside P.T.) (points to the separate table below)

0	0101
1	1100

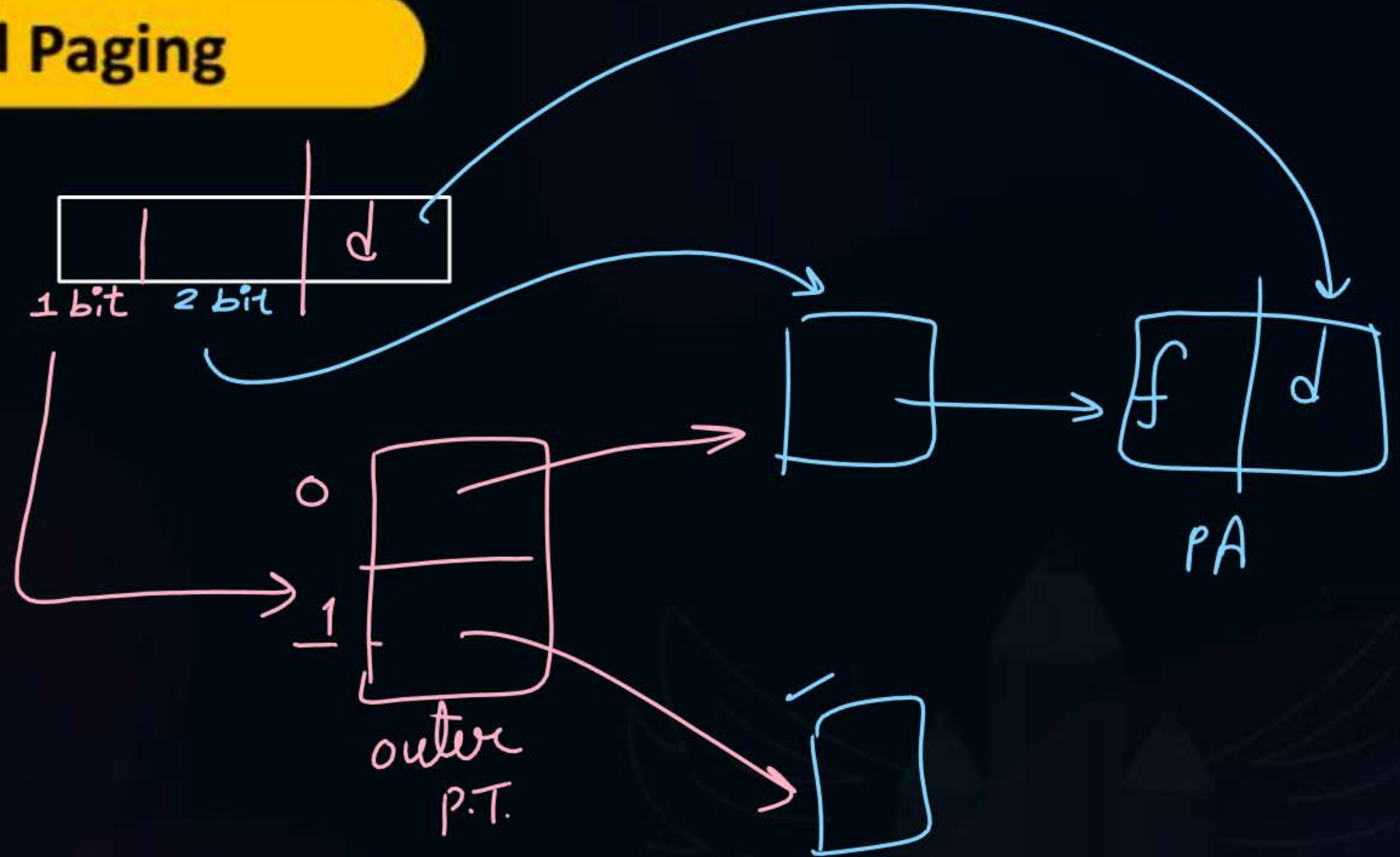
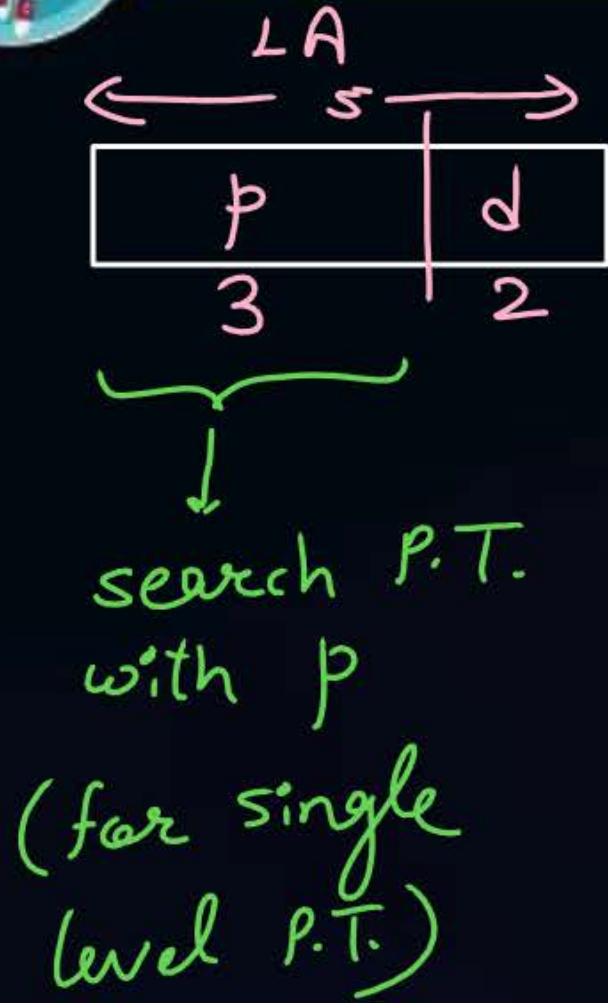
16 frames

Page 0 of PT	mm
0000	
0001	
0010	Page 5
0011	
0100	Page 0
0101	Page 0 of PT
0110	Page 7
0111	Page 1
1000	
1001	Page 4
1010	
1011	Page 3
1100	Page 1 of PT
1101	Page 2
1110	Page 6
1111	

Handwritten notes:
- Page 0 of PT (points to row 0000-0111)
- Page 1 of PT (points to row 1000-1111)
- Red arrows connect PT entries to memory frames: 0100 to 0100, 0110 to 0101, 0110 to 1100.



Topic : Multilevel Paging



ex:-

L.A. =

0	1	0	1	1
---	---	---	---	---

p	d
010	11

	d
0	10
	11

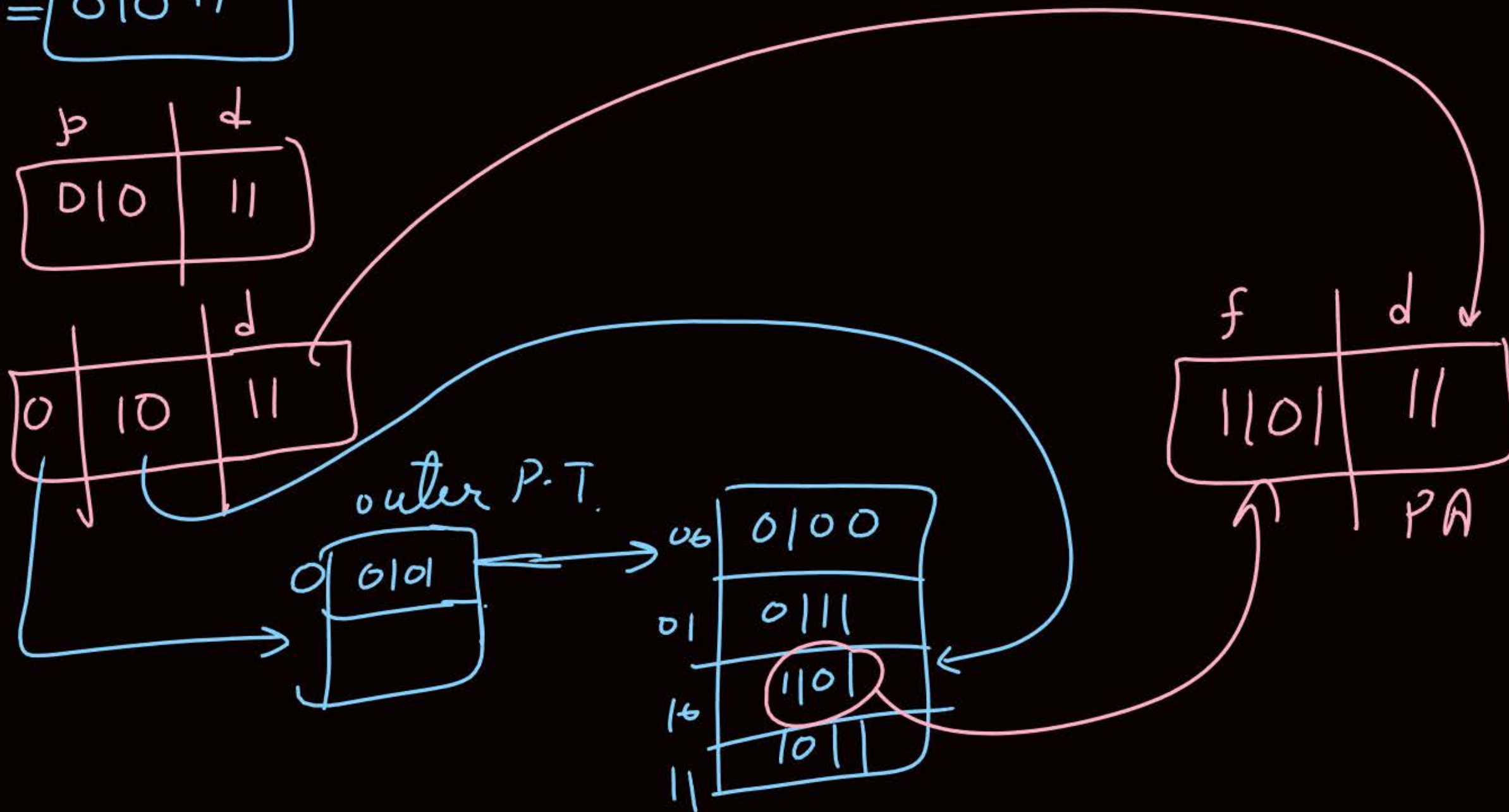
outer P.T.

0	0101
---	------

00	0100
01	0111
10	1101
11	1011

f	d
1101	11

PA



ex:- LA

10101

p	d
101	01

		d
1	0	1

outer PT.

0
1

0101	01
------	----

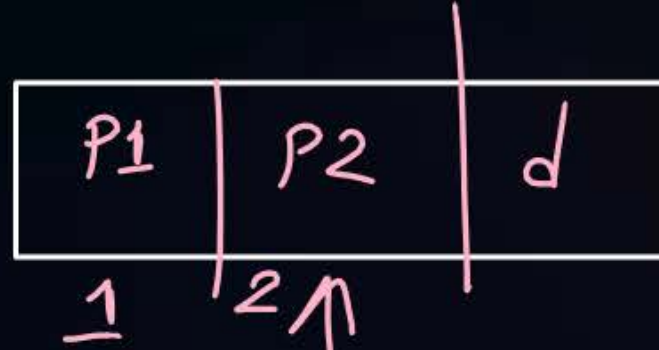
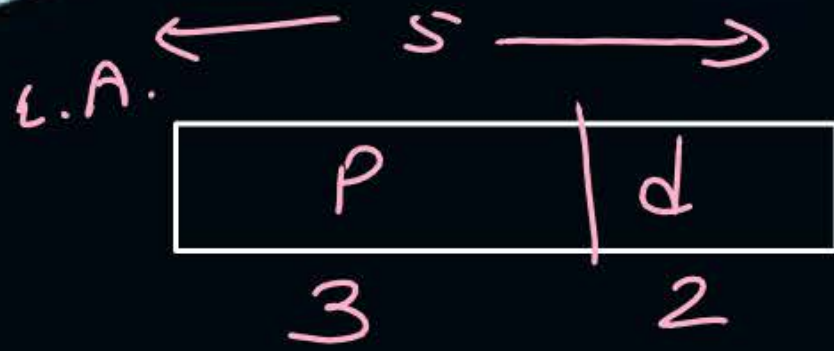
0101	01
------	----

f d

← P.A →



Topic : Multilevel Paging



no. of entries per page = 4
= 2^2

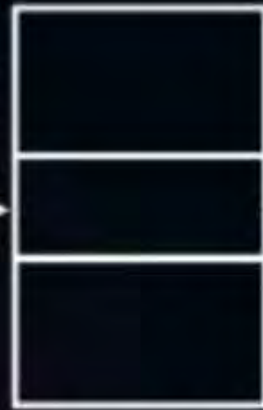


Topic : Multilevel Paging

logical address

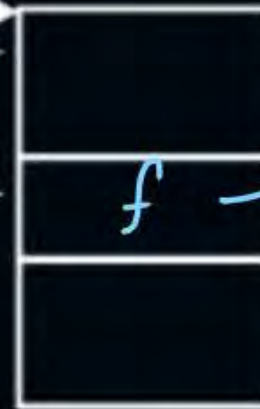


P_1 {



outer -page
table

P_2 {

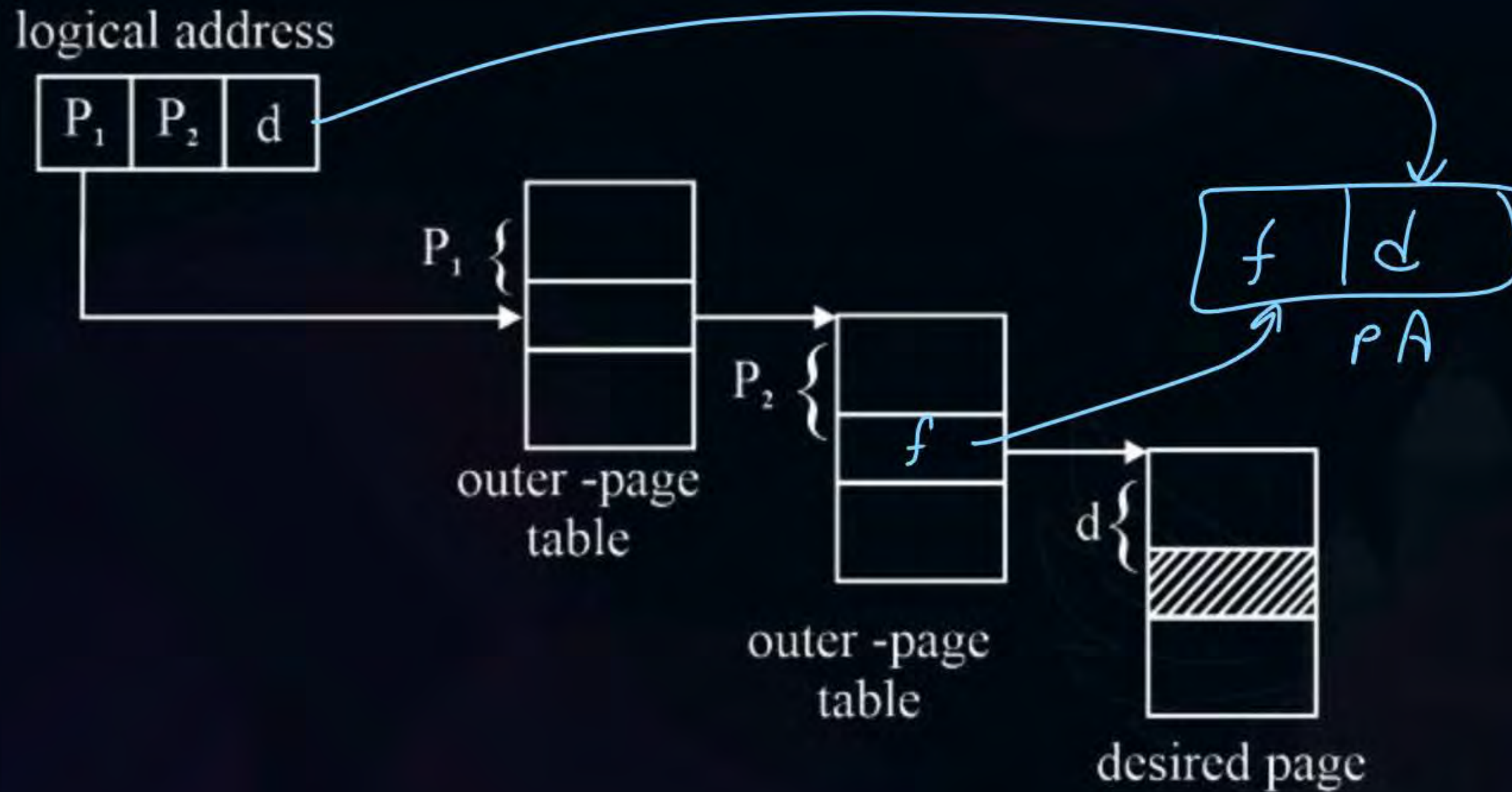
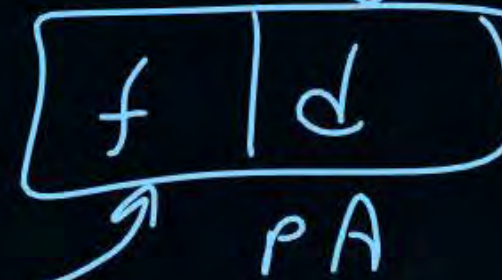


outer -page
table

d {



desired page



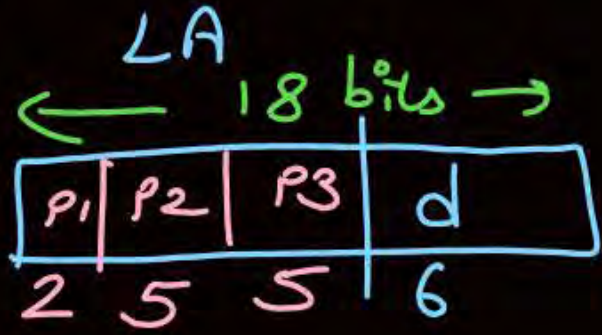


Topic : Multilevel Paging

max no. of bits in $P_2 = \log_2 (\text{no. of P.T. entry stored in page})$
————— || ————— $P_1 = \log_2 (\text{no. of P.T. entry stored in page})$

± If outer P.T. Can not be stored in single page then increase level; until outer P.T. Can fit in single page.

ex:-



Page size = 64B

P.T. entry = 2B

$$\text{no. of entries in Page} = \frac{64B}{2B}$$

$$= 32$$

$$= 2^5$$



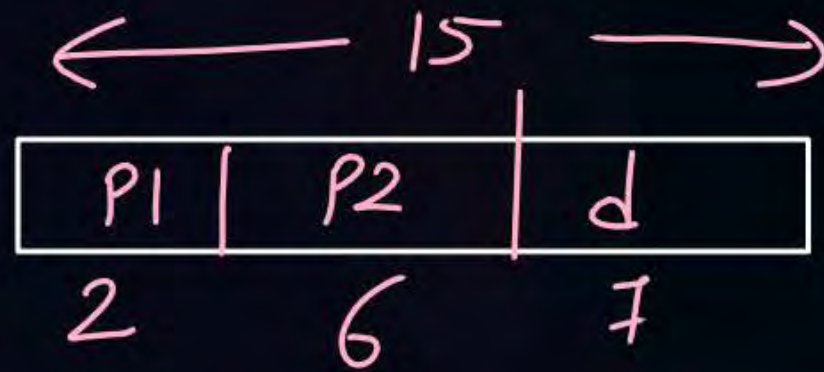
Topic : Question

VA = 15 bits
Page size = 128 bytes
Page table entry size = 2 bytes

$d = 7$

no. of entries per page = $\frac{128B}{2B} = 64 = 2^6$

Number of levels in multilevel page 2?





Topic : Question

VA = 24 bits

Page size = 1Kbytes

Page table entry size = 4 bytes

$d = 10$ bits

} no. of P.T. entries per page = $\frac{1KB}{4B}$

Number of levels in multilevel page 2?

$= 2^8$

24		
P1	P2	d
6	8	10



Topic : Question

VA = 29 bits

Page size = 2Kbytes

Page table entry size = 4 bytes

$$d = 11 \text{ bits}$$

$$\text{no. of P.T. entries per page} = \frac{2 \text{ KB}}{4 \text{ B}}$$

Number of levels in multilevel page 2?

← 29 →		
P_1	P_2	d
9	9	11

$$= \frac{2^{11}}{2^2} = 2^9$$



Topic : Question

VA = 40 bits

Page size = 4Kbytes

Page table entry size = 8 bytes

$$d = 12$$

$$\left. \begin{array}{l} \text{Page size} \\ \text{Page table entry size} \end{array} \right\} \text{no. of P.T. entries per page} = \frac{4KB}{8B}$$

Number of levels in multilevel page 4?

← 40 →

P_1	P_2	P_3	P_4	d
1	9	9	9	12

$$= \frac{2^{12}}{2^3}$$

$$= 2^9$$

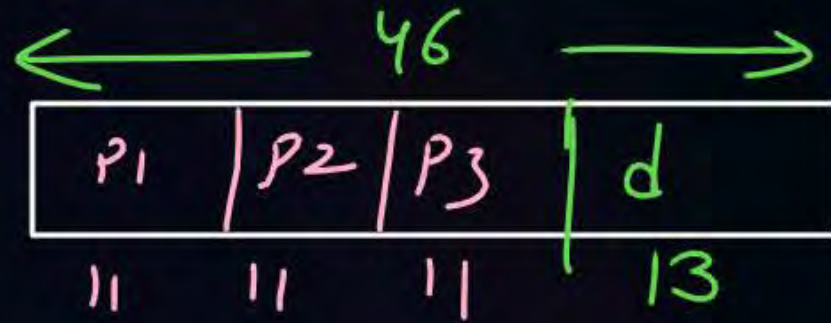


Topic : Question

Ans = 3

#Q. Consider a virtual memory system with physical memory of 8GB, a page size of 8KB and 46-bit virtual address. Assume every page table exactly fits into a single page. If page table entry size is 4B then how many levels of page tables would be required.

$d = 13$ ←



no of P.T entries

$$\text{per page} = \frac{8KB}{4B}$$

$$= 2K$$

$$= 2^{11}$$

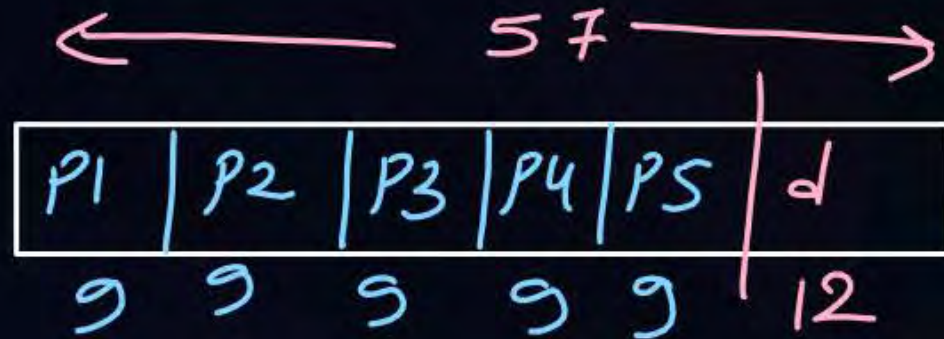


Topic : Question

GATE-2023



- #Q. Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4KB (1KB=1024B) and a page table entry at any of the levels occupies 8 bytes. $\xrightarrow{2^{12}B}$
The value of L is 5?



$$\begin{aligned} \text{no of p.t. entries per page} &= \frac{4KB}{8B} \\ &= 2^9 \end{aligned}$$



2 mins Summary

Topic

Multilevel Paging



Happy Learning

THANK - YOU

